

Contents

1	Consecutive-Prime Sums in Prime Gaps	1
1.1	Abstract	1
1.2	1. Introduction	2
1.2.1	1.1. Motivation	2
1.2.2	1.2. The object	2
1.2.3	1.3. Summary of results and honest positioning	3
1.3	2. Elementary facts	3
1.4	3. The density law	4
1.4.1	3.1. Heuristic derivation	4
1.4.2	3.2. Numerical confirmation	4
1.4.3	3.3. Consequence for the original conjecture	5
1.5	4. Uniqueness and the forbidden-width theorem	5
1.5.1	4.1. Setup and the crude bound	5
1.5.2	4.2. The multiples-of-6 lemma	5
1.5.3	4.3. Width 10	6
1.5.4	4.4. The classification	7
1.6	5. The existence direction	8
1.6.1	5.1. Sub-Poisson emptiness	8
1.6.2	5.2. No deterministic threshold	8
1.6.3	5.3. Why this is hard	9
1.7	6. Discussion	9
1.8	Appendix A. Notation	9
1.9	Appendix B. Reproduction	10
1.10	References	11

1 Consecutive-Prime Sums in Prime Gaps

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1.1 Abstract

Let $p_1 = 2, p_2 = 3, p_3 = 5, \dots$ be the primes and let $S_k = p_k + p_{k+1}$ be the k -th *consecutive-prime sum* (OEIS A001043: 5, 8, 12, 18, 24, 30, ...). We study the distribution of these sums among the prime gaps: given a gap (p_n, p_{n+1}) , how many S_k land strictly inside it?

We establish three things. **(1) A density law:** every S_k with $k \geq 2$ is even, hence composite, hence lands strictly inside a unique prime gap, and the expected number of consecutive-prime sums in a gap of width W near x is $W/(2 \ln(x/2))$. The factor of 2 is the arithmetic content — a consecutive-prime sum lives at the “ $2p$ ” scale, so it is drawn from primes near $x/2$ and is twice as

sparse as the gaps it must fall into. Empirically the mean is ≈ 0.56 sums per gap, so roughly 55% of gaps contain none, 37% contain exactly one, and 8% contain two or more. **(2) A complete finite classification (the main theorem):** the gap widths W for which a gap can *never* contain two consecutive-prime sums are exactly $\{2, 4, 6, 10\}$. The bound rests on the elementary fact that consecutive sums are ≥ 6 apart, combined with a residue analysis modulo 6; width 10 is a genuine straggler, the only forbidden width wide enough to appear capable. **(3) An honest limit:** the “existence” direction — whether every sufficiently wide gap must contain a sum — has no deterministic threshold. Empty gaps persist at arbitrary width relative to the local sum-spacing (we exhibit a width-140 empty gap), and settling the question is of Goldbach/Cramér strength.

All results are verified computationally to $N = 2 \times 10^8$; a self-contained reproduction script is included in Appendix B.

1.2 1. Introduction

1.2.1 1.1. Motivation

Bertrand’s postulate — that there is always a prime between n and $2n$ — suggests that the interval $(p, 2p)$ carries much of the local structure of the primes. This paper follows that intuition to a specific, elementary object. If one adds two consecutive primes, $p_k + p_{k+1}$, the result sits just above $2p_k$; it is a shadow of the pair cast onto the “doubled” scale. Asking where these shadows land among the *larger* primes turns out to have a clean and partly surprising answer.

The starting observation was empirical. Between the consecutive primes 1601 and 1607 there is exactly one number that is a sum of two smaller consecutive primes: $797 + 809 = 1606$. One is tempted to conjecture that this holds throughout — that every gap (excluding the twin-prime gaps, which are too narrow) contains exactly one such sum. That conjecture is *false*, but the way it fails is structured, and the structure is worth recording.

1.2.2 1.2. The object

Definition 1.1. For $k \geq 1$, the k -th **consecutive-prime sum** is

$$S_k = p_k + p_{k+1}.$$

The sequence is $S_1 = 5$, $S_2 = 8$, $S_3 = 12$, $S_4 = 18$, $S_5 = 24$, $S_6 = 30, \dots$ (OEIS A001043).

The sums are strictly increasing, and for a prime gap (p_n, p_{n+1}) we write $W = p_{n+1} - p_n$ for its width and ask for the count

$$c(n) = \#\{k : p_n < S_k < p_{n+1}\}.$$

1.2.3 1.3. Summary of results and honest positioning

The density law of Section 3 is a consequence of the prime number theorem and carries no independent arithmetic surprise; we include it because the constant — exactly 2 — is the quantitative form of the “ p versus $2p$ ” heuristic and because it organizes everything that follows. The **forbidden-width theorem** of Section 4 is the novel contribution: a short, self-contained, and complete classification. The existence discussion of Section 5 is expository; it locates the hard part of the original conjecture inside known deep problems and gives a concrete obstruction.

We claim no deep new mathematics. The interest is in a clean elementary theorem (Section 4) and in the sharp separation between what is provable and what is not (Sections 4 vs. 5).

1.3 2. Elementary facts

Proposition 2.1 (Every sum lands strictly inside a gap). For $k \geq 2$, S_k is even; consequently S_k is composite, and it lies strictly between two consecutive primes — never on a prime.

Proof. For $k \geq 2$, both p_k and p_{k+1} are odd, so S_k is even and > 2 , hence composite. A composite number is not prime, so if $p_m \leq S_k < p_{m+1}$ then $p_m \neq S_k$, giving $p_m < S_k < p_{m+1}$. $\square$

Thus the map $k \mapsto (\text{gap containing } S_k)$ is well defined for $k \geq 2$, and the counts $c(n)$ partition the sums.

Proposition 2.2 (Sums are at least 6 apart, eventually). For k with $p_k \geq 5$,

$$S_{k+1} - S_k = p_{k+2} - p_k \geq 6.$$

Proof. The identity is immediate: $S_{k+1} - S_k = (p_{k+1} + p_{k+2}) - (p_k + p_{k+1}) = p_{k+2} - p_k$. Writing the two gaps $g_k = p_{k+1} - p_k$ and $g_{k+1} = p_{k+2} - p_{k+1}$, we have $p_{k+2} - p_k = g_k + g_{k+1}$, a sum of two consecutive prime gaps. For primes ≥ 5 each gap is even and ≥ 2 ; the only way $g_k = g_{k+1} = 2$ would require three primes $p, p+2, p+4$, but one of any three integers in arithmetic progression with common difference 2 is divisible by 3, so the sole such triple is $(3, 5, 7)$. Hence for $p_k \geq 5$ at least one of g_k, g_{k+1} is ≥ 4 , giving $g_k + g_{k+1} \geq 6$. $\square$

The bound 6 is attained infinitely often in practice (e.g. $5 + 7 = 12$, $7 + 11 = 18$; $11 + 13 = 24$, $13 + 17 = 30$), so Proposition 2.2 is sharp.

1.4 3. The density law

1.4.1 3.1. Heuristic derivation

Fix a location x . A consecutive-prime sum near x has $S_k = p_k + p_{k+1} \approx 2p_k$, so it is generated by a prime $p_k \approx x/2$. As k advances by one, S_k advances by $p_{k+2} - p_k$, which by the prime number theorem is on average twice the local prime gap at $x/2$:

$$S_{k+1} - S_k = p_{k+2} - p_k \approx 2 \ln(x/2).$$

Hence near x the consecutive-prime sums form a sequence of average spacing $2 \ln(x/2)$, i.e. of linear density $1/(2 \ln(x/2))$. The expected number falling in a gap of width W is therefore

$$\boxed{\lambda(W, x) \approx \frac{W}{2 \ln(x/2)}} \quad (3.1)$$

Because prime gaps near x have average width $\ln x$, the mean count per gap is $\approx \ln x / (2 \ln(x/2)) \rightarrow \frac{1}{2}$. This is the quantitative form of the doubling intuition: the sums are drawn from the half-scale $x/2$ and are asymptotically **twice as sparse** as the gaps they must land in, so on average only one gap in two receives a sum.

1.4.2 3.2. Numerical confirmation

Table 1 gives the distribution of $c(n)$ over non-twin gaps with $p_n > 11$, computed to $N = 2 \times 10^7$.

Table 1. Count distribution, non-twin gaps, $p_n > 11$, $N = 2 \times 10^7$ (1,163,197 gaps; mean = 0.561).

$c(n)$	0	1	2	3	≥ 4
gaps	635,687	431,667	73,356	17,455	5,032
fraction	0.546	0.371	0.063	0.015	0.004

The mean is stable across magnitude (0.591 near 10^3 , 0.561 near 10^7), consistent with the slow logarithmic drift in (3.1).

The constant 2 in (3.1) is confirmed directly. Writing the measured linear density of sums near x as $d(x)$ and fitting $c_{\text{fit}} = 1/(d(x) \ln(x/2))$, the value converges to 2 from above:

Table 2. Fitted constant, $N = 5 \times 10^7$.

x	$\sim 10^{2.5}$	$10^{3.5}$	$10^{4.5}$	$10^{5.5}$	$10^{6.5}$	$10^{7.25}$
c_{fit}	2.080	2.052	2.031	2.018	2.014	2.013

Measuring the sum-spacing directly against $2 \ln(x/2)$ gives a ratio converging to 1.007 — a sub-1% lower-order logarithmic correction on top of a clean leading term $2 \ln(x/2)$.

1.4.3 3.3. Consequence for the original conjecture

The naive conjecture — “every non-twin gap after 11 contains exactly one sum” — is false on the existence side: by Table 1 the single most common outcome is an **empty** gap (about 55%). The example $1601 + 1607 \supset 1606$ is a genuine hit, but for a width-6 gap it is only a $\approx 19\%$ event. The corrected statement is the density law (3.1): the count is not 1 but a random-looking quantity with mean $W/(2 \ln(x/2))$.

1.5 4. Uniqueness and the forbidden-width theorem

Although existence fails, a sharpened form of the conjecture — *when a gap does contain a sum, it contains only one* — is nearly true and, for narrow gaps, exactly true. Among the 1,242,066 non-empty non-twin gaps below 5×10^7 , 81.7% contain exactly one sum; the 18.3% with two or more are concentrated in the wide gaps. We now determine precisely which widths *force* uniqueness.

1.5.1 4.1. Setup and the crude bound

Suppose a gap $(a, a+W)$ contains two consecutive-prime sums $S_k < S_{k+1}$. Both lie in an open interval of length W , so their separation satisfies $S_{k+1} - S_k \leq W - 2$ (the sums are even and a is odd, so the extreme interior even integers are $a + 1$ and $a + W - 1$, spanning $W - 2$). By Proposition 2.2, for a large enough that the generating primes exceed 5 (which holds for every gap of width ≥ 8 ; the finitely many tiny gaps are checked directly), the separation is also ≥ 6 . Hence a double requires $W - 2 \geq 6$, i.e.

$$W \geq 8. \tag{4.1}$$

In particular, **any gap of width ≤ 6 contains at most one consecutive-prime sum**. This already covers the twin gaps ($W = 2$) and widths 4, 6.

1.5.2 4.2. The multiples-of-6 lemma

The tight case $S_{k+1} - S_k = 6$ has rigid arithmetic.

Lemma 4.1. If $S_{k+1} - S_k = 6$ (with $p_k \geq 5$), then $S_k \equiv S_{k+1} \equiv 0 \pmod{6}$.

Proof. Separation 6 means the two prime gaps (g_k, g_{k+1}) are $(2, 4)$ or $(4, 2)$.

- Case (2, 4): the triple is $q, q+2, q+6$. Requiring none divisible by 3 forces $q \equiv 2 \pmod{3}$. Then $S_k = 2q+2 \equiv 0$ and $S_{k+1} = 2q+8 \equiv 0 \pmod{3}$.
- Case (4, 2): the triple is $q, q+4, q+6$, forcing $q \equiv 1 \pmod{3}$, whence $S_k = 2q+4 \equiv 0$ and $S_{k+1} = 2q+10 \equiv 0 \pmod{3}$.

In both cases the sums are $\equiv 0 \pmod{3}$, and being even they are $\equiv 0 \pmod{6}$.
 $\square$

So a tight double consists of **two multiples of 6, six apart**. Whether a given width can accommodate this reduces to counting the multiples of 6 interior to the gap — and that count is forced by $W \pmod{3}$, because the gap's lower endpoint has a forced residue.

Lemma 4.2 (Endpoint residue). Let $(a, a+W)$ be a prime gap with $a > 3$. - If $W \equiv 1 \pmod{3}$ then $a \equiv 1 \pmod{6}$. - If $W \equiv 2 \pmod{3}$ then $a \equiv 5 \pmod{6}$. - If $W \equiv 0 \pmod{3}$ then $a \equiv 1$ or $5 \pmod{6}$.

Proof. Both a and $a+W$ are primes > 3 , so neither is $\equiv 0 \pmod{3}$. If $W \equiv 1$, then $a \equiv 2 \pmod{3}$ would give $a+W \equiv 0$, impossible; so $a \equiv 1 \pmod{3}$. If $W \equiv 2$, then $a \equiv 1$ would give $a+W \equiv 0$; so $a \equiv 2 \pmod{3}$. Combining with a odd (so $a \not\equiv 3 \pmod{6}$) yields the stated residues mod 6. $\square$

Counting multiples of 6 in the open interval $(a, a+W)$ under Lemma 4.2 gives the **slot count** $s(W)$ = the maximum number of multiples of 6 that can lie interior, which is the number of admissible positions for tight-double members:

Table 3. Slot count $s(W)$ = interior multiples of 6.

W	2	4	6	8	10	12	14	16	18	20
$W \pmod{3}$	2	1	0	2	1	0	2	1	0	2
$s(W)$	1	0	1	2	1	2	3	2	3	4

A tight double needs $s(W) \geq 2$. This first happens at $W = 8$, **fails again at** $W = 10$, and holds for all $W \geq 12$ (the offset pattern only adds slots as W grows). The lone dip at $W = 10$ is the crux.

1.5.3 4.3. Width 10

Proposition 4.3. No prime gap of width 10 contains two consecutive-prime sums.

Proof. Let $(a, a+10)$ be a prime gap. By Lemma 4.2, $a \equiv 1 \pmod{6}$; every such gap has $a \geq 139$, so the generating primes exceed 5 and Proposition 2.2 applies. A double would have separation $\sigma \in \{6, 8\}$ (even, ≥ 6 by Prop. 2.2, $\leq W-2=8$).

Separation 6. By Lemma 4.1 both sums are multiples of 6. With $a \equiv 1 \pmod{6}$, the only interior multiple of 6 is $a+5$ ($a-1 \leq a$ and $a+11 \geq a+10$ are excluded). One slot cannot hold two members. Impossible.

Separation 8. The only two interior even integers 8 apart are $a + 1$ and $a + 9$. With $a \equiv 1 \pmod{3}$ their residues are $a + 1 \equiv 2$ and $a + 9 \equiv 1 \pmod{3}$, so a double here would need $(S_k, S_{k+1}) \equiv (2, 1) \pmod{3}$. But a separation-8 triple is $(2, 6)$ or $(6, 2)$ (the pattern $(4, 4)$ gives $q, q+4, q+8 \equiv q, q+1, q+2$, always hitting a multiple of 3, so it is not a prime triple for $q > 3$): - $(2, 6)$: $q \equiv 2 \pmod{3}$, giving $(S_k, S_{k+1}) \equiv (0, 2)$; - $(6, 2)$: $q \equiv 2 \pmod{3}$, giving $(S_k, S_{k+1}) \equiv (1, 0)$.

Neither is $(2, 1)$. Impossible. $\square$

1.5.4 4.4. The classification

Theorem 4.4 (Forbidden widths).

- (a) If $W \in \{2, 4, 6, 10\}$, then no prime gap of width W contains two consecutive-prime sums.
- (b) For every even $W \geq 8$ with $W \notin \{10\}$, prime gaps of width W containing two consecutive-prime sums exist.

Proof. (a) Widths 2, 4, 6 are handled by (4.1); width 10 is Proposition 4.3. (b) For $W = 8$ and every even $12 \leq W \leq 60$ an explicit example is exhibited computationally (Table 4 lists the first; the search to $N = 2 \times 10^8$ found doubles at every such width). Structurally, $s(W) \geq 2$ for all these W by Table 3, so the required pair of interior multiple-of-6 positions exists, and prime triples realizing a $(2, 4)$ or $(4, 2)$ pattern at the half-scale populate them. $\square$

Remark 4.5. The forbidden set is exactly $\{2, 4, 6, 10\}$: no forbidden width exceeds 10. Part (b) is verified for $8 \leq W \leq 60$; its extension to all even $W \geq 12$ is expected on the same slot-counting basis, and the *infinitude* of doubles at each such width would follow from the prime k -tuple conjecture (the relevant triples are admissible constellations). The elementary content — part (a), the complete list of widths that force uniqueness — is unconditional.

Table 4. Doubles by width, $N = 2 \times 10^8$. “first a ” is the lower endpoint of the first gap of that width containing two sums.

W	gaps	with ≥ 2	rate	first a
2	813,370	0	0	—
4	813,714	0	0	—
6	1,429,538	0	0	—
8	624,743	1,181	0.19%	389
10	805,277	0	0	—
12	1,017,047	7,332	0.72%	199
14	556,481	11,593	2.08%	317
16	411,449	4,581	1.11%	1,933
18	739,453	22,016	2.98%	523
30	443,840	62,707	14.13%	4,831
60	62,576	34,355	54.90%	43,331

The first genuine double occurs in the width-8 gap (389, 397): it contains both $390 = 193 + 197$ and $396 = 197 + 199$.

Corollary 4.6 (Sharpened conjecture). For gaps of width $W \in \{2, 4, 6, 10\}$, the original “only one” property holds without exception: such a gap contains at most one consecutive-prime sum. For all other widths it can fail, with failure rate increasing in W .

1.6 5. The existence direction

The remaining half of the original conjecture — does every *sufficiently wide* gap contain at least one sum? — behaves very differently. It has no elementary resolution, and we explain why.

1.6.1 5.1. Sub-Poisson emptiness

If the sums fell as independent random points of density $1/(2\ln(x/2))$, the fraction of empty gaps would be $e^{-\lambda}$ with λ as in (3.1). The measured empty fraction is markedly *smaller* and the discrepancy grows with λ :

Table 5. Empty fraction vs. Poisson prediction, $N = 10^8$.

λ	0.375	0.625	0.875	1.125	1.625	2.125
$P(\text{empty})$	0.659	0.448	0.289	0.183	0.065	0.022
$e^{-\lambda}$	0.687	0.535	0.417	0.325	0.197	0.119

The empty fraction decays faster than $e^{-\lambda}$ — roughly $e^{-2.3\lambda}$ — because consecutive-prime sums are **under-dispersed**: they march with spacing tightly concentrated near $2\ln(x/2)$ rather than clustering like Poisson points, and an evenly spaced sequence leaves fewer empty intervals. So wide gaps are non-empty *more* reliably than a random model predicts; the original intuition is, if anything, better than chance would justify.

1.6.2 5.2. No deterministic threshold

Nevertheless there is no width — however large relative to the local sum-spacing — that guarantees a hit. Empty gaps persist out to $\lambda \approx 4$.

Example 5.1. The gap (79,092,617, 79,092,757) has width 140 and contains **no** consecutive-prime sum, although $\lambda \approx 4.0$ (the width is four times the local sum-spacing). The sum sequence steps $\dots, 79,092,612, 79,092,758, \dots$ — a single stride of 146 that straddles the entire gap. This is a double coincidence: the primes near $x \approx 79 \times 10^6$ are locally sparse (the wide gap) and the primes near $x/2 \approx 39.5 \times 10^6$ that generate the nearby sums are simultaneously sparse at the aligned location.

1.6.3 5.3. Why this is hard

Proving that every gap wider than some explicit function of x contains a sum would require ruling out coincidences of the type in Example 5.1 — a controlled statement about the joint behavior of prime gaps at x and prime spacing at $x/2$. This couples the distribution of gaps to itself across scales and is of the same order of difficulty as Goldbach- and Cramér-type problems. We therefore leave existence as:

Open Problem 5.2. Is there a function $f(x) = o(x)$ such that every prime gap (p_n, p_{n+1}) with $p_{n+1} - p_n \geq f(p_n)$ contains a consecutive-prime sum? Equivalently, bound the largest empty gap at height x .

1.7 6. Discussion

The trajectory of the original conjecture is a compact case study. A pattern noticed at a single gap (1601, 1607) generalized in three ways with three different epistemic statuses:

1. **Existence (“always one”)** — **false**, replaced by the density law (3.1) whose only content is the doubling constant 2, a restatement of the prime number theorem at the half-scale.
2. **Uniqueness (“at most one”)** — **a theorem for exactly the widths** $\{2, 4, 6, 10\}$ (Theorem 4.4), unconditional and elementary, with width 10 the single non-obvious member.
3. **Wide-gap existence** — **open and hard** (Open Problem 5.2), provably not settleable by width alone (Example 5.1).

Within the Factor Skyline program the consecutive-prime sum is the simplest object that lives one dyadic scale above the primes that define it, making the interaction between a gap at x and the prime density at $x/2$ explicit and elementary. The clean separation here — an unconditional finite classification sitting directly beside a Goldbach-strength open problem — mirrors the template/escape split elsewhere in the framework: the part fixed by congruences is provable, the part requiring cross-scale coincidence control is not.

1.8 Appendix A. Notation

- p_k : the k -th prime, $p_1 = 2$.
- $S_k = p_k + p_{k+1}$: the k -th consecutive-prime sum (OEIS A001043).
- $W = p_{n+1} - p_n$: gap width. A *twin gap* has $W = 2$.
- $c(n) = \#\{k : p_n < S_k < p_{n+1}\}$: sums in gap n .
- $\lambda(W, x) = W/(2 \ln(x/2))$: expected count (density law).
- $s(W)$: slot count, interior multiples of 6 (Table 3).

1.9 Appendix B. Reproduction

All tables were produced by the following self-contained script (Python 3, standard library only). Sieve to N , form the consecutive-prime sums, sweep the gaps with a two-pointer, and tabulate.

```
import math
from collections import defaultdict

def sieve(n):
    bs = bytearray([1]) * (n + 1); bs[0] = bs[1] = 0
    i = 2
    while i * i <= n:
        if bs[i]: bs[i*i::i] = bytearray(len(bs[i*i::i]))
        i += 1
    return [i for i in range(n + 1) if bs[i]]

def analyze(N):
    P = sieve(N)
    S = [P[k] + P[k+1] for k in range(len(P) - 1)] # consecutive-prime sums
    maxS = S[-1]
    tot = defaultdict(int); dbl = defaultdict(int); first = {}
    dist = defaultdict(int) # count-distribution, non-twin, p_n>
    j = 0
    for n in range(len(P) - 1):
        a, b = P[n], P[n+1]
        if b >= maxS: break
        if a < 3: continue # skip the degenerate width-1 gap (2,3)
        while j < len(S) and S[j] <= a: j += 1 # first sum > a
        jj = j; c = 0
        while jj < len(S) and S[jj] < b: c += 1; jj += 1
        w = b - a
        tot[w] += 1
        if c >= 2:
            dbl[w] += 1
            first.setdefault(w, a)
        if a > 11 and w > 2:
            dist[c] += 1
    # forbidden widths = those with zero doubles (verify == {2,4,6,10})
    forbidden = sorted(w for w in tot if dbl[w] == 0)
    mean = sum(k * v for k, v in dist.items()) / sum(dist.values())
    return dist, forbidden, first, mean

if __name__ == "__main__":
    dist, forbidden, first, mean = analyze(2 * 10**8)
    print("non-twin (p_n>11) count distribution:", dict(sorted(dist.items())))
```

```

print("mean sums/gap:", round(mean, 3))
print("forbidden widths (zero doubles):", forbidden)
print("first double at width 8 / 12:", first.get(8), first.get(12))

```

Expected output (abridged): forbidden widths [2, 4, 6, 10]; mean ≈ 0.56 ;
first width-8 double at $a = 389$.

1.10 References

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